PICKTACULAR

Operating Room Pick-list Decision Support System for Cambridge Memorial Hospital

University of Waterloo | Department of Management Science and Engineering

Cambridge Memorial Hospital

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Problem Analysis

Context

Same surgery → Different pick-lists
→ Unnecessary variation in inventory

Why It Matters

↑ Costs — More SKUs, more waste
↑ Complexity — Harder to manage + prepare OR
↓ Visibility — No easy comparison across surgeons

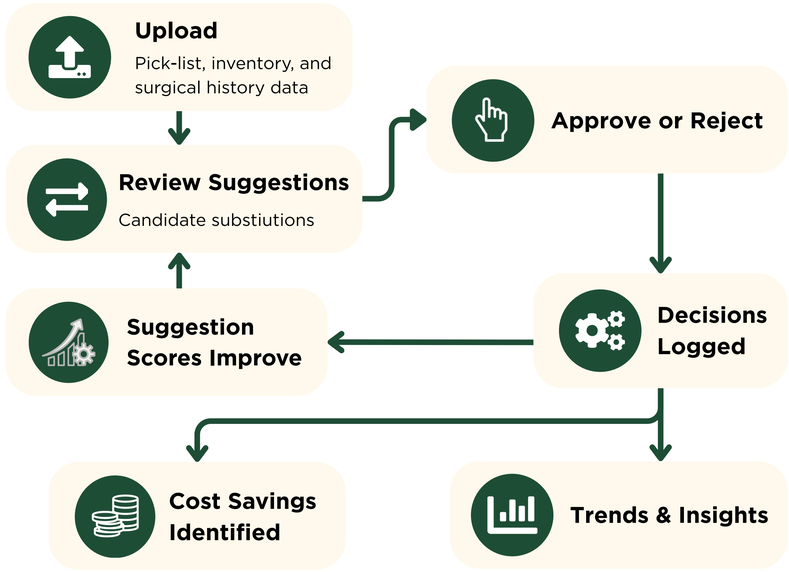
What is at Stake

Unsolved → Excess inventory, ongoing inefficiency
Solved → Streamlined SKUs, better decisions

Objective

Identify variation → Suggest consolidation
→ Support better inventory decisions

User Flow



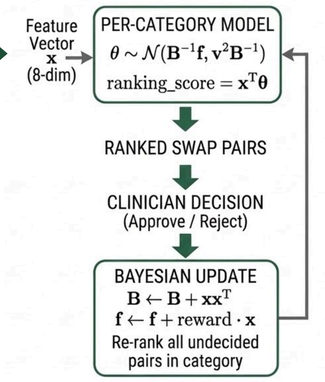
Adaptive Learning

1. Feature Engineering

Feature	Signal
Overlap Coefficient	Text Similarity
Prefix Match	Item Type Match
Sizing Match	Size Constraint
Number Mismatch	Size Compatibility
Shared Numbers	Specification Overlap
Price Ratio	Cost Similarity
Total Surgeons	Usage Breadth
Co-occurrence	Usage Overlap*

7 features selected via greedy forward selection against a LLM-labelled ground truth

2. Thompson Sampling Loop



Per-category models learn which features predict interchangeability

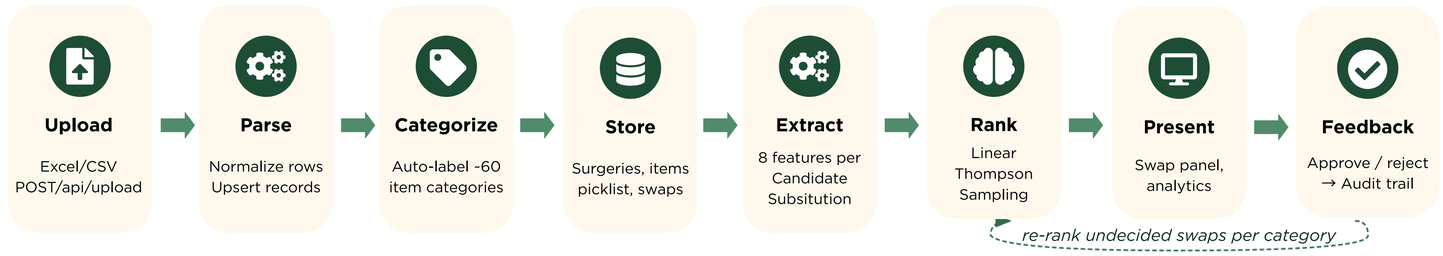
Each decision immediately re-ranks all undecided pairs

Thompson Sampling balances exploration vs. exploitation

Software Structure

Frontend	Backend	Database
React + Vite SPA	FastAPI (Python)	SQLite (Local)

Data Flow



Dashboard Features

Surgery Overview
→ Browse surgeries, compare surgeons, identify variation hotspots

Interchangeable item suggestions
→ Approve, reject, or override swap candidates

Analytics & Savings
→ Track holding costs and consolidation efforts over time

Activity Log
→ Every user action is recorded with a justification, all reversible

Verification & Validation

Stakeholder Walkthrough
→ Bi-weekly evaluations on early prototype w/ CMH

Survey on Baseline vs. Ground Truth
→ Compare algorithm suggestions against CMH approve/reject decisions

Usability Testing
→ Improve usability of dashboard interface

Potential Impacts

Reduced SKU count
Simplified inventory planning

Cost reduction

Waste reduction

Impacts are potential because this is a **decision-support**, not decision-making system.

Patient outcomes remain as the **first priority**.